

A Low-Cost First Test of a Brain-Boosting Idea in Rats

Introduction in simple words · Researcher: Solano Moreira Furlan

What is the idea?

Scientists found that a natural substance in the fluid around young brains, called FGF17, seems to help old mice remember things better. Another substance, called 7,8-DHF, is thought to help brain cells make stronger connections with each other. We want to know: if we give both, and also give the animals a rich, stimulating place to learn, do they become better at solving different kinds of problems than normal rats of the same age?

What will we do?

We will use a small number of healthy young rats (64 in the main test, 130 counting helper animals) in four groups: normal rats; rats living in the learning environment; rats given the two substances; and rats given the two substances and the learning environment. Later we test all of them on three different tasks — a puzzle where the rules suddenly change, a maze with new problems, and a swimming-pool memory test. Before the main test, a small first step checks whether the substances switch on their expected signal in the rat brain (this does not prove they reach every brain cell). If that first signal is missing, we stop early and save money. Some scientists also doubt that 7,8-DHF works the way people first thought, so we treat that as an open question. The rats get a small brain operation, daily injections, limited food on some days and swimming tests, and are humanely put to sleep at the end; an animal-ethics committee must approve everything first.

Who does the work?

A good laboratory in India will run the whole experiment, using this document together with detailed lab procedures that the lab staff write and the researcher approves. The researcher guides it from far away and keeps the secret group codes, so the staff giving the medicines do not know which rat got what (they can see which rats live in the enriched space), and the researcher checks the results independently.

How much will it cost?

The first step and the main experiment together are planned to cost about USD 9,700 for materials only (supplies, animals, and chemicals). It could be as low as 8,500 or as high as 11,600. Staff pay and the lab's overhead charges are NOT included, and together they could be a large part of the real bill; the Indian lab must quote them. A second part (looking at the brain tissue under the microscope) costs about USD 1,700 more and can wait.

What will this test tell us — and not tell us?

This is only a first, small test. It can tell us whether the experiment is practical, whether the substances reach the brain cells, and whether the results lean in a promising direction. It cannot prove that the idea works, and it cannot say anything about people.

What if it goes well?

Then we plan a bigger, stronger experiment. Its first part would separate what each ingredient does. Its second part would use optogenetics — a new technique that switches chosen brain cells on and off with light — to see how brain activity interacts with these substances.

The long-term dream

In the far future we hope this kind of approach might help people who have learning difficulties or problems with how brain cells connect, and older people with Alzheimer's disease, by helping the brain rebuild connections. That is a hope, not a result. Many steps, long safety testing, and strict ethics rules come first, and there are real obstacles (for example, how to get these substances into a human brain safely). The rest of this document gives the technical details.

Technical protocol (India, core ≈ USD 10,000)

Does a candidate plasticity-promoting treatment (FGF17 + 7,8-DHF) combined with a structured learning environment raise performance across different cognitive tasks in rats? Staged feasibility study run by an Indian host laboratory under remote oversight.

Researcher / sponsor	Solano Moreira Furlan (remote; the host laboratory runs the experiment)
Document	India initial-test protocol, version 5 (rounds 1–4 applied; rounds 5–6 pending)
Date	9 October 2026
Status	Not approved by an ethics committee. Every reviewer verdict so far: “approve with changes”. Prices are estimates, not quotes. “crf” in the request was read as CSF (assumption to confirm).

Executive summary (technical)

Question. Does a candidate plasticity-promoting treatment — FGF17 (identified in young cerebrospinal fluid; Iram 2022 [1]) plus 7,8-DHF (a putative TrkB/BDNF-pathway agonist whose mechanism is disputed [2–4]) — combined with a structured enriched learning environment (ELE) raise the performance of healthy young rats across three tasks, one of which (set-shifting) uses stimulus dimensions not practised in the ELE, compared with untreated peers?

Design. 64 rats in the main 2×2 (≈10–11 completers per arm); 130 rats in total including 42 engagement satellites. Staged: Phase 0a (engagement and reagents) must pass before Phase 0b (behaviour); Phase 0c (EdU/Olig2 histology) is a deferred add-on on banked tissue. It informs, and does not confirm, whether a larger study is justified.

Budget (honest). Consumables-only core (0a + 0b, tissue banked) ≈ USD 9.7k base; ≈ 8.5k optimistic; ≈ 11.6k pessimistic. Phase 0c adds ≈ 1.7k. Personnel, service fees and institutional overhead are not included and must be quoted by the host; the all-in cost is unknown until then.

1. Aim, hypotheses and what the data can claim

- **Primary aim (feasibility):** estimate completion rate, variance and direction of performance in a 2x2 of drug package × environment; establish reagent activity and pathway engagement.
- **Exploratory hypothesis:** D+E (drug package + enrichment) performs better than control peers (C) and than either component alone.
- **Transfer:** set-shifting is the primary transfer readout (odour and digging-medium dimensions are excluded from the ELE). Hebb-Williams and the water maze are spatial; the ELE has tunnels, climbing and wheels, so they are flagged as spatial, not transfer.
- **Allowed claims:** feasibility, attrition, variance, direction with 95% interval of D+E – C (overall index; set-shifting reported separately), pathway engagement (ERK), descriptive structure if Phase 0c is run — for the drug package as a whole.
- **Not allowed:** “smarter”, “general intelligence” or a g factor [8,9]; neuroplasticity; single-agent claims; TrkB-specific claims (pERK shows pathway engagement only); human or disease claims; replication of Iram 2022 (aged-mouse rescue; this asks about enhancement in healthy young adults, which no study found has tested).

2. Long-term goal, next experiment (including optogenetics) and translational roadmap

Recommended wording (translational reviewer, adopted):

“Our long-term aim is to test whether combining a plasticity-promoting biological treatment (an FGF17-type factor plus a validated TrkB/BDNF-pathway agent) with structured enriched learning can produce measurable cognitive benefit in humans. We pursue two **separate** programmes with distinct biology, models, ages and ethics: (a) neurodevelopmental conditions with altered connectivity (learning disabilities), and (b) age-related cognitive decline and early Alzheimer’s disease. We hypothesise that supporting synaptic and myelin plasticity is **one possible contributor** to such benefit; this is a hypothesis to be demonstrated with structural and functional evidence, not an assumption, and it is not addressed by the present study. Prior neurotrophin programmes in Alzheimer’s have not shown benefit (including a direct-delivery AAV-NGF trial), brain delivery of protein factors is a major obstacle, and both candidate agents face unresolved questions (human CNS delivery of FGF17; whether 7,8-DHF engages TrkB). Each stage proceeds only on pre-registered target-engagement, structural and safety gates.”

Stage	Content	Gate to proceed
0 – this study	0a engagement/reagents → 0b behaviour in healthy young rats → 0c histology (deferred add-on).	Section 8.
1a – next experiment: powered rat confirmation	Both sexes; full factorial separating FGF17, 7,8-DHF and enrichment (and exercise); sustained infusion by pump; dose-response; durable vs on-drug effects; TrkB and FGFR necessity tests; synaptic readouts (PSD-95/synaptophysin on banked tissue, LTP).	Replicated effect on a pre-registered primary; structural effect.
1b – next experiment: optogenetics (only if 1a is positive; mechanistic, not translational)	Question: how does controlled neuronal activity interact with FGF17 and with 7,8-DHF-driven plasticity? Candidate designs (to be designed and reviewed in their own round): (a) activity × plasticity-opening — optogenetic stimulation of a defined circuit (e.g. CaMKIIa-driven channelrhodopsin in prefrontal or hippocampal pyramidal neurons) during or after learning, with and without FGF17/7,8-DHF, asking whether activity amplifies oligodendrocyte-precursor proliferation, myelination and synaptic plasticity (mouse precedent: premotor-cortex stimulation, Gibson 2014 [10]; extension to prefrontal/hippocampal circuits is an extrapolation); (b) necessity — inhibitory opsins during learning or consolidation; (c) cell-type targeting of oligodendrocyte-lineage cells needs Cre lines that are mostly mouse (species switch or an AAV-based rat strategy). Requirements: AAV vectors with institutional biosafety (IBSC) approval (referral to RCGM only if the risk category requires), fibre implants, calibrated light sources, opsin-negative/fluorophore-only and light-only controls (heating, visual artefacts), histological verification of opsin and fibre placement, and monitoring for stimulation-induced seizures/kindling. Beyond this budget; own funding, equipment, expertise and ethics approval.	1a positive; opsin tools validated in the host lab.
2 – disease models (separate tracks)	Aged rodents (Iram-type rescue); Alzheimer’s models (5xFAD, APP/PS1, TgF344-AD) — amyloid-model rescue has repeatedly failed to translate; neurodevelopmental models at appropriate ages: Fmr1 knockout (fragile X) and Down-syndrome models (Ts65Dn is being superseded) as separate tracks.	Benefit with engagement and structural evidence.

Stage	Content	Gate to proceed
3 – drug development	Clinically deliverable form (intranasal, intrathecal or engineered/receptor-selective agent — unproven); a validated TrkB-pathway drug with human-relevant PK; two-species toxicology; FGF-class risks (mitogenesis/tumour promotion, angiogenesis); regulatory plan (CDSCO NDCT Rules 2019; ICMR ethics; US/EU IND).	Delivery solved; clean toxicology.
4 – human studies	Phase 1 safety/PK; biomarkers (MRI myelin water fraction/DTI, EEG, cognitive tests); structured learning input as digital training; Phase 2 in MCI/early AD and, separately, in neurodevelopmental populations (DCGI scrutiny; assent plus proxy consent; independent data-safety monitoring).	Safety and biomarker signal.

Hard barriers that must be stated: no validated way exists to deliver FGF17 to the human brain; the mitogenic and angiogenic risk of an FGF-class CNS agent may never clear a paediatric risk-benefit assessment for a non-fatal indication such as a learning disability, and enhancement in people without disease carries a higher ethical bar than treatment of disease; 7,8-DHF is reported not to activate TrkB reliably in cell assays [2–4]; no human trials of FGF17 or 7,8-DHF were found (verify); healthy-animal enhancement may not transfer to disease or humans. The FGF23/phosphate effect is not the relevant class risk (a different, endocrine FGF).

What this study must deliver to earn Stage 1a: reagent activity and FGF17 pathway engagement (0a); a positive direction on the overall index (0b), with set-shifting reported separately; and, if Phase 0c is run, an EdU+/Olig2+ direction. Covariation of structure with behaviour cannot be judged at this n.

3. Design and animals

Young adult male Wistar rats (about 8–10 weeks at arrival) from a CPCSEA-registered source (the regulator is now also called CCSEA; ICAR-IVRI Izatnagar INR 350; NISER INR 300–500 plus transport; NIN's animal unit is being reorganised — confirm supply directly). Males only. Housed on a reversed 12:12 light cycle so that dark-phase work falls in working hours. The sponsor randomises by baseline weight and source litter and holds the allocation and drug-code key (section 10); **each wave contains 8 rats per arm**; coded syringes/vials; surgeon, injector and behaviour operators blind to drug. The environment arm cannot be fully blinded, so ELE handlers and test-day scorers are different people, all behaviour is video-recorded under coded IDs and scored blind, 20% are re-scored for inter-rater agreement (ICC), and apparatus is cleaned between rats. Session order across arms follows a Latin square, so arm is not confounded with time of day.

Group	Content	Enrolled
Main cohort	4 arms × 16: C (vehicle + standard), E (vehicle + ELE), D (FGF17 + 7,8-DHF + standard), D+E; 2 waves × 8 per arm	64
Engagement satellite – FGF17	Vehicle / 15 / 30 / 60 min after an ICV bolus, 6 each; rapid dissection and snap-freezing of subventricular (SVZ/subependymal) tissue and hippocampus; pERK/total ERK	24
Engagement satellite – DHF	Vehicle / 45 min / 120 min after the 7th daily dose (6 each); pERK/total ERK (descriptive)	18
Partner rats	12 unoperated partners (6 for ELE arenas, 6 for control arenas; never swapped; aggression pre-screened; ≈5 rats per partner per condition)	12
Practice/pilot	6 cannula/dye placement and EdU labelling checks (incl. storage-time control); 6 vehicle-tolerability checks (3 low-DMSO solution vs 3 CMC-Na suspension, 14 days, necropsy scoring)	12
Total	Cannulated rats: 64 + 24 + 6 = 94	130

Enrolment is 16 per arm because ICV-cannula attrition in rats is commonly 30–40%; the plan assumes 62–69% completion (10–11 per arm). Cannula loss is logged by arm. Because climbing, wheels and partners exist only in the enrichment arms, cap loss is expected to be higher there: protected recessed caps with mesh ceilings are used, an arm-stratified tipping-point analysis is pre-specified, the delivered FGF17 dose is reported per D+E completer, and a completion difference of >25 percentage points between arms triggers a review (and possible stop). **Welfare:** the combined burden (cannula for ≈10 weeks, 53 daily injections, food restriction, swimming) is classified as moderate severity, with a scoresheet and humane endpoints (>15% weight loss, weight below 85% of the growth-adjusted target, injection-site lesions, cannula infection, hypothermia rule); the IAEC will be asked for a pre-submission review of severity. Pre-specified exclusions: dye/histology-confirmed ventricular placement, humane endpoints, swim speed <50% of cohort median. Housing: pair-housed with a perforated divider if the IAEC permits, otherwise single-housed, identically in all arms.

Power (honest). With N=40–44 completers an interaction contrast reaches 80% power only for $f \approx 0.43$ – 0.46 , and a pairwise D+E vs C comparison only for $d \approx 1.3$ (unadjusted); power at $f = 0.25$ is 0.34–0.37. A cross-species enrichment meta-analysis reports $g \approx 0.44$ in rodents (an unpublished dissertation [15]), and healthy young rats are near ceiling on simple tasks, so a null result is likely even if a modest true effect exists. Pilot effect sizes must not be used to power the next experiment from the point estimate [12,13].

4. Interventions

Component	Specification
FGF17 (days 0–6)	Recombinant mouse FGF17 (mouse–rat sequence alignment documented before ordering). Order one 25 µg vial first, bioassay it, then order the rest (second import, 6–12 weeks, is on the critical path). Chronic guide cannula to the right lateral ventricle (scaled coordinates). Four boluses on days 0, 2, 4, 6: 3 µl of ~100 µg/ml (0.3 µg each; 1.2 µg per rat) in aCSF + 0.1% BSA + heparin 10 µg/ml, awake, 0.5–1 µl/min with the injector left 2–3 min; aliquots thawed fresh each session. Supply: ≈44 µg for the main and satellite rats plus bioassay and dead volume against 75 µg ordered (a 4th vial is the contingency). Vehicle boluses in all other rats. <u>Delivery gap:</u> Iram 2022 infused for about a week by osmotic pump (Alzet 1007D ≈84 µl); four boluses total 12 µl, ≈7× less volume and pulsatile. The delivered dose was not retrievable; the Methods (open copy at the Weizmann/PMC/Nature sites) must be read before IAEC submission, and until then FGF17 is exploratory and a null is confounded with delivery.
7,8-DHF	5 mg/kg IP daily, days 0–52 (53 injections; rotated sites). Daily order: testing (if scheduled) → 7,8-DHF injection → ELE/control session 30 min later, so the agent is on board during enrichment (mouse half-life ≈2 h) and the next day's testing falls at trough (≈18–24 h after dosing). Testing and sessions occur in a fixed 4-h window; arm order within the window follows a Latin square. Rat precedent: 7 days to ≥5 weeks at 5 mg/kg (a rotenone-model study; one optic-nerve study followed rats to 60 days but protection was lost by then), with vehicles from 1% to 30% DMSO [5]; 53 days is at the long end of precedent. Vehicle (decided by the 14-day tolerability check, 6 rats): default is a low-DMSO solution (about 5% DMSO / 40% PEG300 / 5% Tween-80 / saline, ≈2.5 mg/ml, a vendor-listed formulation); fallback is an aqueous CMC-Na suspension (no peer-reviewed intraperitoneal PK found; an IP suspension of an insoluble compound may give erratic, depot-like absorption, so a null could reflect exposure and a plasma/brain bridging check is then required). Oil vehicles are avoided (repeated intraperitoneal oil, especially peanut or mineral, can cause granulomatous inflammation; mild with corn or olive oil in a 5-day mouse study). Necropsy scoring of peritoneal adhesions/granulomas in all main-cohort rats; identical vehicle in all arms.

Component	Specification
Enriched learning environment (ELE)	Days 0–52, one 60-min pair session/day (test rat + partner) 30 min after the 7,8-DHF injection, in the dark phase under red light [6,7]. This is a weak enrichment dose (declared). SOP with photos to be completed by the host before IAEC submission (arena size, object list ≥ 3 new objects/day, latch/string-pull/lever/chew puzzle devices on a six-level ladder, running wheels, tunnels/climbing, small fixed-calorie rewards deducted from the daily ration; engagement minutes and ladder progression logged). <u>Excluded from the ELE: odour discrimination and digging puzzles.</u> Control arena: same partner, handling and duration, empty.
EdU (banked label)	Start at 50 mg/kg IP on days 2, 4, 6 (labelling and a stored-tissue control tested in the practice group). Staining is Phase 0c.

5. Timeline and workload (day 0 = first bolus; waves run sequentially)

Days	Activity
–21 to –15	Arrival and 7-day acclimatisation/quarantine; reversed light cycle.
–14 to –11	Cannula surgery with analgesia (≤ 8 rats/day); ≥ 10 days recovery before day 0.
–10 to –1	Recovery, handling, arena habituation, weighing.
0–6	FGF17/vehicle boluses (0, 2, 4, 6); EdU (2, 4, 6).
0–52	ELE/control sessions and 7,8-DHF/vehicle every day (53 doses), weekends included.
14–21	Gradual food restriction to 90–95% of the growth-adjusted target; held through day 45; humane limit $< 85\%$; daily ration at a fixed clock time 2 h after the last session.
22–36	Task 1: set-shifting (parameters in Appendix A). Training to criterion (cap 7 days), staggered; tests ≤ 4 rats/day with two operators (2–4 h per rat).
37–45	Task 2: Hebb-Williams — adaptation days 37–39, then 6 chosen problems, one per day (days 40–45), fixed order, 6–8 trials per problem; error zones defined in advance; split-half reliability reported. Food restriction ends on the evening of day 45.
46–52	Task 3 (last; non-appetitive): water maze — acquisition days 46–50, probe day 51, visible-platform test day 52 (Appendix A).
53–56	Under anaesthesia an ICV dye injection (3 μ l) confirms patency, then decapitation; a 2-mm coronal slab through the cannula track is cut fresh from the right hemisphere and photographed for placement, and the rest is snap-frozen. 8 rats/day balanced by arm and time of day; per-animal handling ≤ 12 min (ice-cold freeze side first). Hemisection: the non-cannulated (left) hemisphere is cut into 1.5–2 mm coronal slabs into 4% PFA for 48 h (counts restricted to the outer 1 mm of each slab, pre-registered), cryoprotected and banked, with a fixation-gradient check on a deep unlabelled section; the remainder of the cannulated (right) hemisphere is snap-frozen and banked at -80°C (dorsal hippocampus and corpus callosum dissected on ice clear of the cannula track, or documented if not; no phospho urgency).

Waves. The two waves are strictly sequential for arena use and surgery: wave 2's day 0 is wave 1's day 70, so wave 2's arrival (its day –21 = wave 1 day 49) overlaps only wave 1's water maze, and wave 2's surgery (its days –14 to –11 = wave 1 days 56–59) follows wave 1's euthanasia. Each wave needs, daily, 32 pair-sessions (16 ELE + 16 control) in 4 arenas = 32 arena-hours (8 h per arena, within the 12-h dark phase), 32 injections and weighing, plus two testing operators on test days: **four trained operators, a surgeon and a weekend rota** (time-budget table in Appendix B). In-life per wave 77 days elapsed (day –21 to day 56); total ≈ 21 weeks (wave 1 day –21 to wave 2 day 56 = 147 days). The fixed task order (set-shifting $\rightarrow$ Hebb-Williams $\rightarrow$ water maze) is declared; ELE after testing and DHF are a post-training consolidation window, declared.

6. Endpoints and tissue

Primary analysis (per task). The three task scores are standardised with the grand mean and the pooled within-arm SD estimated in-sample (no prior data exist), winsorised on residuals at ± 3 , signs aligned. They are analysed in one linear mixed model (task as a repeated factor; random intercept per animal; task-specific residual variances; arm $\times$ task terms; wave as a fixed effect; baseline weight as the only covariate), using all animals with ≥ 1 task, with Kenward-Roger intervals. The gate statistic is the **index contrast** (mean of the three z-scores, $D+E - C$) in per-task-SD units; set-shifting alone and the index without the water maze are reported separately. Sensitivity analyses: complete cases; unstructured residual covariance; tipping-point (δ -adjusted) analysis for informative dropout (the mixed model is valid under missing-at-random only). Wheel running is zero in two arms and post-restriction weight is post-randomisation, so neither is a primary covariate. Completion by arm and task is reported with an imbalance flag (> 20 percentage points). The inter-task correlation matrix is computed within arm from residuals and is descriptive.

Set-shifting score. Criterion 6 consecutive correct, maximum 60 trials per stage; score = $\log(\text{trials to criterion at the extradimensional shift, capped at } 60)$; rats that do not reach it are assigned the cap; the primary score is the raw log extradimensional-shift score (conditioning on the post-randomisation intradimensional-shift score, and rank/censored analyses, are sensitivity analyses only). **Water maze:** mean proximity to the platform in the 24-h probe [19]; quadrant time also reported.

Block	Readouts	n
Behaviour	Three tasks; blinded video scoring.	all completers
Banked tissue (every main-cohort animal)	Left-hemisphere slabs (1.5–2 mm; counts restricted to the outer 1 mm, pre-registered) for EdU/Olig2 ($\pm$ later CC1; $\pm$ FluoroMyelin or Luxol Fast Blue counterstain as a cheap structural readout; corpus callosum primary, dentate/CA1 secondary; counts valid only within matched depth; immersion fixation gives lower, depth-dependent quality than perfusion — declared); right hemisphere frozen (PSD-95/synaptophysin and other westerns deferred to Stage 1a or a funded add-on). Co-stored EdU+ and no-EdU control blocks; stored cold in buffer with 1–2% formaldehyde (cryoprotected sections at -20°C) and capped at 12 months; chromogenic EdU if fluorescence background is high.	$\approx 10/\text{arm}$
Engagement (satellites, snap-frozen)	pERK/total ERK in SVZ/subependymal tissue (FGF17 gate anchor) and hippocampus; ice-cold dissection (<2 min), phosphatase inhibitors (NaF, orthovanadate, β -glycerophosphate), FGF2-stimulated positive-control lysate and pre-specified assay CV on every blot; FGF17 at 15, 30 and 60 min; 7,8-DHF at 45 and 120 min after the 7th daily dose (descriptive); every gel carries a bridging control sample and the FGF2-stimulated lysate so fold-changes compare across gels (the 42 satellite samples cannot share one gel).	6 per group

7. Statistics and pre-registration

- Register on OSF before surgery (sponsor does this); ARRIVE 2.0 [14]; flow diagram of enrolled, excluded and completed animals.
- Estimation only: unadjusted contrasts with Kenward-Roger 95% intervals as primary; an animal-level (cluster) bootstrap that re-runs standardisation, winsorising and the model in every replicate, stratified by arm only, is reported as sensitivity (BCa is unstable at $n \approx 10$); no Holm adjustment because no significance claims are made. Interaction exploratory.
- Partner identity is described only (6 partners per environment cannot support a random effect). Wave (calendar, reagent lots, operator learning) is confounded; arm is balanced across waves so the main contrast is protected, but arm $\times$ wave cannot be assessed with two waves.
- Sizing of the next study: use the upper 80% confidence limit of the pilot's raw-unit residual SD per task [18] and the full factorial/interaction design; the “64 per arm for 0.5 SD” figure applies to a two-group comparison only.

8. Pre-registered decision rules and operating characteristics

Phase	Pass condition			
0a – reagents	FGF17 bioassay active on the first vial; mouse-rat alignment documented; DHF suspension stable.			
0a – FGF17 engagement (gate)	SVZ/subependymal pERK/total ERK ≥ 1.5 -fold vehicle at ≥ 1 of 15/30/60 min (n=6 per group; acceptance CV pre-specified; positive-control lysate on every blot). FGF17 signals through FGFR3/FGFR4 on oligodendrocyte-lineage cells [16]; receptor expression in adult SVZ and the time points are assumptions anchored only by the FGF2 control curve. Simulation (n=6 per group, best of three time points): false-pass $\approx 3\%$ at assay CV 30% and $\approx 18\%$ at CV 50%; true-pass $\approx 75\%$ if the true fold-change is $1.5\times$, $\geq 97\%$ at $2\times$. A borderline result (best fold 1.3–1.5) triggers one pre-specified re-test (6 more per group at the best time point). If not met, the main cohort does not start; test a higher dose or continuous infusion in a small follow-up.			
0a – DHF engagement	Reported descriptively, not a gate (n=6, assay CV likely exceeds the effect). If not met, the DHF component is reported as “effect of unestablished mechanism”.			
0b – feasibility	$\geq 50\%$ of the 64 enrolled rats (≥ 32) complete all three tasks; cannula placement $\geq 75\%$ correct. (Binomial P($\geq 32/64$): 0.82 at 55%, 0.96 at 60%, 0.995 at 65% true completion; with clustering by wave and surgeon about 0.68, 0.80 and 0.89.)			
0b – direction	Index contrast D+E – C $\geq +0.4$ per-task SD (illustrative conversion assuming one common inter-task correlation: +0.69 index-SD at r = 0, +0.55 at r = 0.3, +0.47 at r = 0.6) <u>and</u> D+E $\geq$ both E and D in point estimate; interval reported, not gated.			
Provisional go	Phase 0a passed AND 0b feasibility AND 0b direction.			
0c (deferred add-on) – confirmed go	EdU+/Olig2+ density in corpus callosum higher in D+E than C (direction). If 0c is not run, the decision stays provisional.			
True situation (n=10/arm; index contrast)	r=0	r=0.3	r=0.6	r=0.3 + EdU direction*
No effect	5%	8%	10%	4%
Modest (E 0.3, D 0.1, D+E 0.4)	39%	37%	36%	32%
Target (E 0.3, D 0.2, D+E 0.6)	70%	62%	57%	53%
Additive (E 0.3, D 0.3, D+E 0.6)	68%	59%	54%	51%

True situation (n=10/arm; index contrast)	r=0	r=0.3	r=0.6	r=0.3 + EdU direction*
Large (E 0.5, D 0.3, D+E 1.0)	96%	91%	85%	78%
Enrichment alone (E 0.5, D+E 0.5)	40%	38%	38%	19%

*EdU direction simulated as independent with a true drug effect of 0.5 SD where a drug effect exists (the “modest” row has a smaller drug effect, so its EdU column is not directly comparable). Other checks: with 8 completers per arm the false-go rate under no effect is $\approx 10\%$ and the target-scenario go rate $\approx 59\%$; the index without the water maze has a $\approx 9\text{--}10\%$ false-go rate. The rule is lenient and **cannot separate synergy from additivity** (62% vs 59%); it only screens for a promising direction. A “go” justifies the larger study only; a “no-go” identifies the failure mode (delivery, engagement, ceiling, attrition) and is not evidence that the approach cannot work.

9. India logistics and budget

Item	What the review found
Ethics	Rats are small animals: IAEC approval is sufficient (no central CPCSEA/CCSEA forwarding). The IAEC must meet at least every 6 months (quarterly for larger programmes); proposals are due ≈ 30 days ahead; ask for an interim review. The IAEC principal investigator must be host staff.
Imports	FGF17 and antibodies are imported in at least two shipments (vial-first FGF17 order; later orders): 6–12 weeks each, customs-house-agent fees ($\approx$ INR 5–15k per consignment), dry-ice courier. Notification 51/96 gives $\approx 5\%$ duty on “scientific instruments, apparatus, equipment and spare parts” for DSIR-registered importers; coverage of reagents and any GST exemption (2017 notifications 45/2017-CT(R), 47/2017-IGST, against a head-of-institution certificate) are unconfirmed and the importing institute itself must hold DSIR registration. Without it, landed cost is $\approx 1.3\text{--}1.5 \times$ list. Duty/GST rates are reviewers’ knowledge, not verified.
Equipment access (precondition)	The budget assumes written, free access to: stereotaxic frame, water maze + tracking software, set-shift and maze apparatus, cryostat, fluorescence microscope and stereology software, western blot system. Hourly facility charges are NOT in any scenario; without written access the budget is void.
Personnel, services, overhead	Not in the budget. Four operators, a surgeon and a coordinator for $\approx 6\text{--}7$ months including Phase 0a, plus institutional overhead (typically 15–30%, institute-specific), must be quoted by the host. The USD 10k figure is consumables-only.
Timeline	IAEC 1–6 months; procurement 2–3 months in parallel (the second FGF17 import, 6–12 weeks, follows the bioassay and is on the critical path); Phase 0a 1–2; in-life ≈ 4.8 months (21 weeks); analysis 1–2. About 8–16 months from IAEC submission to core results; host search, contracting and FCRA/GST arrangements come before that and are not counted.

Budget (USD; list prices from search plus reviewer knowledge; to be replaced by quotes):

Line (core = Phase 0a + 0b)		Base	Notes
Rats: 130 incl. satellites, partners, pilots, with transport		1.0k	INR 300–500 each
Housing/care ≈ 74 days per main-cohort rat; partners longer		1.5k	per-diem not found; optimistic 1.0k
FGF17 3 $\times$ 25 μ g, landed		1.0k	Cohesion $\approx$ USD 480/50 μ g via an aggregator; R&D; 25 μ g mouse vial price not found; PeproTech lists human only; 4th vial 0.45k in pessimistic
7,8-DHF ≈ 4 g, landed		0.35k	TCI $\approx$ USD 62/g
Cannulas, dummies, screws, cement (94 rats)		0.5k	
Anaesthesia, analgesia, surgical and terminal consumables (≈ 94 cannulations, ≈ 130 terminal procedures)		1.0k	
ELE arena, wheels, objects, puzzle devices (18% GST on fabrication)		0.65k	
Behaviour consumables and fabrication (set-shift pots/media/odours, Hebb-Williams maze) incl. GST		0.8k	
EdU (banks the label)		0.2k	
ERK block, landed: pERK, total ERK, HRP secondaries, loading control, FGF2 positive control		1.55k	list ≈ 1.15 k
Western reagents (membranes, ECL, inhibitors)		0.45k	
Two shipments with customs-agent fees		0.6k	
IAEC fee and miscellaneous		0.1k	
Core base total, consumables only, no reserve		9.7k	excludes personnel, overhead, equipment fees
Phase 0c add-on (deferred): EdU click kit + Olig2 + secondary landed 1.34k, IHC reagents 0.25k, shipping 0.1k		1.7k	CC1 ($\approx$ USD 594 list) and PSD-95/synaptophysin westerns are further add-ons
Full programme base		11.4k	
Scenario	Core	Full	Assumptions (itemised)
Optimistic	≈ 8.5 k	≈ 9.8 k	minus housing 0.5k, FGF17 0.15k, antibodies at list 0.4k, one cheaper shipment 0.2k (core); 0c at list ≈ 1.35 k
Base	≈ 9.7 k	≈ 11.4 k	as above
Pessimistic	≈ 11.6 k	≈ 13.6 k	15% overrun plus a 4th FGF17 vial; equipment fees, personnel and overhead still excluded (the base already assumes no DSIR concession)

Phases: 0a ≈ 3.8 k (satellite rats, first FGF17 vial, ERK antibodies, surgery, first shipment) with a stop rule at that spend; 0b ≈ 5.9 k; 0c ≈ 1.7 k deferred. **First cuts if prices run high:** enrol 14 per arm (waves of 7 + 7; feasibility gate becomes ≥ 28); drop

the 60-min satellite group; run 0c later. The western block must not be dropped (it carries the only gate). **Must not be cut:** dye/histology placement check, FGF17 bioassay and alignment, blinding/randomisation, DHF tolerability check, visible-platform control, banking of tissue.

10. Lab handover and remote oversight

The researcher will not be on site; the host laboratory runs the experiment. This PDF is the working protocol; the following must exist before animals arrive.

Item	Responsibility / requirement
Legal form (decide first)	Either (a) an academic collaboration under an MoU — then the institute needs FCRA registration or prior permission and a designated account for foreign contributions (likely required for foreign research funds, received in a designated bank account; payment for services rendered is outside FCRA; confirm in writing with the institute's accounts office; the researcher's nationality is stated in the agreement because it may change the treatment); or (b) a fee-for-service contract — invoice, GST position or export-of-services status and TDS must be agreed. The choice determines payment mechanics and is a precondition for submission.
Host selection and due diligence	Criteria: CCSEA/CPCSEA-registered establishment with a functioning IAEC; written record of ICV cannulation, set-shifting and water-maze work; access to all listed equipment; a named attending veterinarian; references from two previous external collaborations. Trade-off: a CRO offers fee-for-service and quality systems but rarely these assays; an academic lab offers expertise but expects authorship and slower timelines.
Payment tied to gates	Tranches: (1) at signature (set-up, animal order); (2) after the host-competence gate; (3) after Phase 0a passes; (4) at data lock. A failed 0a stops spending at its tranche. The agreement states who bears currency risk.
Governance and welfare oversight	Protocol amendments need the host PI, the sponsor and (where required) the IAEC; a named attending veterinarian can stop the study; weekly remote welfare report (weights, scores, cap losses, adverse events); escalation: sponsor within 48 h, institute head within 7 days; working language English; a fixed weekly call slot overlapping both time zones.
Quality standard	Non-GLP; the host works to GLP-like good practice (contemporaneous records, calibrated equipment, training records, retained raw data).
Roles and accountability	Host principal investigator (accountable for IAEC compliance, animal welfare; also the IAEC PI); study coordinator (one named person); four trained operators plus a surgeon and a weekend rota; sponsor (design, randomisation, drug-code key, registration, independent analysis).
Agreement terms	Funding, overhead and payment schedule; who pays for overruns and for a failed Phase 0a; IP ownership; data and tissue ownership; insurance and liability; stop rules; publication and authorship (ICMJE criteria); data sharing.
Host-competence gate (before Phase 0a)	Practice-group evidence: $\geq 80\%$ successful ICV cannulation by dye; western blot CV from the lab's previous work; set-shifting and water-maze pilot records; verified equipment list. Sponsor decides go/no-go on this evidence.
SOP pack (host completes with photos; sponsor reviews)	Surgery and cannula care; ICV bolus injection; vehicle preparation and dosing; ELE and control sessions; set-shifting; Hebb-Williams; water maze incl. temperature/hypothermia rules; food restriction and weighing; euthanasia, hemisection, slab fixation and tissue banking; satellite dissection and western protocol; per-day time-budget table; welfare scoresheet; Appendix A parameters signed off.
Blinding and randomisation	Sponsor generates the allocation, labels coded vials and arm IDs, and keeps the key; host staff handle coded IDs only; unblinding after data lock.
Data integrity and monitoring	Bound or electronic lab notebooks with audit trail; raw files hashed on upload; protocol deviations classed minor/major with major ones reported within 48 h and a sponsor decision; scheduled live-video audits and source-data verification of $\approx 10\%$ of records; weekly summary (enrolment, weights, exclusions, adverse events, cap losses) and photo/video check-in; raw data and images shared with the sponsor for independent blinded analysis.
Tissue and image transfer	Banked tissue stays at the host unless a material transfer agreement and the necessary export permits are in place; image format and storage (coded IDs, backup) agreed in advance.
Reagent chain	Quotes agreed; vial-first FGF17 order; bioassay result and alignment shared before the main cohort starts; cold-chain receipt log.
Decisions	After 0a and 0b the sponsor issues a written decision (provisional go / stop / redesign) before further spending.

Appendix A. Task parameters (defaults; host to confirm and sign off before animals arrive)

Task	Parameters
Set-shifting (digging; Birrell & Brown 2000; Tait et al. 2018 [11,17])	Two ceramic pots per trial; stimulus dimensions: digging medium and odour (neither is used in the ELE); 2–3 days digging habituation to a baited pot; example exemplar sets (host to fix ≥ 6 each before animals arrive): odours — cinnamon, nutmeg, clove, cumin, thyme, vanilla; media — sawdust, shredded paper, sand, plastic beads, wood chips, cotton; stages: simple discrimination, compound discrimination, intradimensional shift, extradimensional shift (reversal stages omitted to limit session length; this weakens specificity of the extradimensional shift — declared); criterion 6 consecutive correct; maximum 60 trials per stage; 3-min trial limit; error on first incorrect dig; bait $\approx \frac{1}{2}$ cereal piece; pot positions pseudo-random; training cap 7 days.
Hebb-Williams (Rabinovitch & Rosvold 1951 [11])	Closed-field maze with barrier layouts; 3 days adaptation with food reward including the standard practice problems; 6 test problems chosen from the standard 12 by host and sponsor before animals arrive (fixed order, same layouts for all rats; investigator-chosen subset, not validated); 6–8 trials per problem, one problem per day; error zones defined in advance; split-half reliability reported.
Morris water maze	Pool ≈ 150 – 180 cm, 25 ± 1 °C, hidden 10-cm platform 1.5–2 cm below the surface in opaque water (non-toxic white paint), one pool, ≥ 4 distal cues, dim white light (≈ 10 – 20 lux, measured; spectrum recorded); 4 trials/day, 60 s maximum, four quasi-random start points, ≥ 5 min between trials in a warm cage (32 °C); core temperature before and after sessions (rest day if < 34.5 °C); 5 acquisition days; 60-s probe 24 h after the last acquisition day; visible-platform test (4 trials) after the probe; scores: mean proximity (Gallagher), target-quadrant time, swim speed, floating, thigmotaxis.
ELE / control session	Dark phase under red light; pair session of 60 min (test rat + partner); 4 arenas; Latin-square order across arms; log object list, puzzle level, rewards, engagement minutes, wheel revolutions.

Appendix B. Operator time budget per wave (to be confirmed by the host)

Period	Daily work	Operator-hours/day
Days –14 to –11	Cannula surgery, 8 rats/day (surgeon + assistant)	≈ 16
Days 0–6	ICV boluses and EdU (≈ 2 h) on bolus days, plus the daily routine below	≈ 22 – 24
Days 0–52, every day incl. weekends	ELE/control: 32 pair-sessions in 4 arenas = 8 h per arena, 1 operator per 2 arenas ≈ 16 h; DHF injections ≈ 1.5 h; weighing, ration and cage checks ≈ 1.5 h; cleaning ≈ 1 h	≈ 20
Test days (set-shift)	4 rats $\times$ 2–4 h with two operators, on top of the daily routine	≈ 32
Days 53–56	Euthanasia, 8 rats/day $\times$ ≤ 12 min, two operators, on top of the routine	≈ 24
Weekends	Daily routine only	≈ 20

Four operators at 8 h cover about 32 operator-hours a day (peak); weekend cover needs a rota. The surgeon is additional on surgery days.

11. Validation record for this adaptation

Round	Reviewers	Verdicts	Main issues → fix
Earlier A	3 Opus (design; pharmacology; logistics)	All approve with changes	Pump ends before testing; pump cost; middle-aged rats; tissue conflict → boluses, young adults, new sourcing.
Earlier B	3 Opus + 1 Haiku; the aim was changed	All approve with changes	Transfer overlap; social confound; composite not a g factor; attrition; timeline; budget.
Translational	1 Opus	Approve with changes	Overclaiming; neurotrophin history; FGF safety; paediatric ethics.
Round 1	5 Opus + 2 Sonnet	All 7 approve with changes	Gate miscalibrated; BrdU/Olig2 on adjacent sections; phospho pre-analytics; timelines; ELE trained set-shift dimensions; budget overhead; 25 inconsistencies.
Round 2	5 Opus + 2 Sonnet	All 7 approve with changes	“Go” rule not a gate; aliased covariates; complete-case bias; no frozen tissue; n=4 satellites; transfer mislabelled; Hebb-Williams reliability; wave workload; shipping vs vial-first; 17 inconsistencies.
Round 3	5 Opus + 2 Sonnet	All 7 approve with changes; 8 objections called blocking (all fixable on paper)	Whole-hemisphere immersion fixation fails for deep regions → slab fixation, contralateral hemisphere; corn-oil IP ×53 → CMC-Na default + necropsy scoring; overlapping waves impossible in 4 arenas → sequential waves; <7 days surgical recovery → surgery D-14; no staff/overhead costing; legal form, IAEC PI, IP and insurance; data-integrity regime; host-competence gate; wave as fixed effect; set-shift scoring; restriction vs humane endpoint; time-of-day; parameter appendix; SD conversion error; 8 inconsistencies → all applied in v4.
Round 4	5 Opus + 2 Sonnet (one Opus review ended on a safeguard flag after delivering its report)	All approve with changes; blocking: cannula placement vs frozen hemisphere, host executability, gate-linked payments, CMC-Na exposure, FGF17 gate anchoring, ED-on-ID overadjustment, gate power, ELE-correlated cap loss	Dye + fresh slab for placement; low-DMSO solution default and extra DHF time point; FGF17 receptor rationale, gate power and re-test; raw log-ED primary; protected caps and attrition stop; DHF before ELE; Appendix B time budget; host selection, payment tranches, governance; 1.5-2 mm slabs; cross-gel bridging control; in-life 21 weeks; 130 rats.
Rounds 5-6	5 Opus + 2 Sonnet each	(pending)	Two further rounds requested by the researcher.

12. Items the reviewers still do not like / remaining gaps (after round 3)

#	Gap	Kind
1	Bolus FGF17 is not the Iram infusion ($\approx 7\times$ less volume); dose unanchored until the Methods are read; null confounded with delivery.	Pumps in the larger study
2	Power: only very large effects detectable; healthy rats near ceiling; the direction rule is lenient and cannot separate synergy from additivity (38% go for enrichment alone).	Intrinsic to a pilot
3	7,8-DHF may be inert in vivo; 53 injections at the long end of precedent; pERK shows pathway engagement only; DHF gate descriptive.	Declared
4	ELE confounds social contact, exercise, puzzles and spatial experience; weak 60-min dose; arm cannot be fully blinded; set-shifting is the only clean transfer readout.	Declared
5	Cumulative welfare burden may be challenged by the IAEC; attrition may exceed the plan; staffing (4 operators, weekends) to be confirmed by the host.	Pre-review; host
6	Prices unverified; personnel, overhead and equipment fees excluded; pessimistic core 11.6k exceeds the $\approx$ USD 10k target; DSIR reagent coverage and FCRA/GST status unconfirmed.	Quotes and written confirmations
7	Structure (EdU/Olig2) is deferred, so the provisional go lacks a structural marker; immersion-fixed deep-region histology is lower quality than perfusion; no synaptic or neuroplasticity readout in this study.	Phase 0c / Stage 1a
8	Optogenetics (Stage 1b) and the human roadmap are aspirations without design detail or validated tools in the host lab.	Own review round
9	Literature checks used search snippets only; Iram’s delivered dose still not retrieved; human-trial and neurotrophin statements to be verified.	Researcher/host to verify
10	Vehicle choice may confound exposure (suspension fallback); 7,8-DHF is at trough at each day’s testing; wave and calendar are confounded (arm $\times$ wave not estimable).	Declared

#	Gap	Kind
11	Host executability depends on SOPs, quotes and written confirmations (FCRA/GST, DSIR, equipment access) that are not yet in hand.	Host to supply

13. References and verification status

Checks were by search snippets only (PubMed, PMC, Crossref and publisher pages unreachable). “Verified” means consistent across indexed records.

#	Citation	Status
1	Iram T, Kern F, Kaur A, et al. Young CSF restores oligodendrogenesis and memory in aged mice via Fgf17. <i>Nature</i> 2022;605:509–515. doi:10.1038/s41586-022-04722-0 (an Author Correction exists)	Verified; delivered dose not retrieved
2	Boltaev U, et al. Multiplex quantitative assays indicate a need for reevaluating reported small-molecule TrkB agonists. <i>Sci Signal</i> 2017;10(493). doi:10.1126/scisignal.aal1670	Verified
3	Todd D, Gowers I, Dowler SJ, et al. <i>PLoS One</i> 2014;9(2):e87923. doi:10.1371/journal.pone.0087923	Citation verified; DHF content via reviews
4	Do small molecules activate the TrkB receptor in the same manner as BDNF? <i>Pharmaceuticals</i> 2021;14(8):704 (authors not captured).	Verified (review)
5	Rawlings-Mortimer F, et al. 7,8-DHF during water-maze training in mice. <i>Front Syst Neurosci</i> 2023. doi:10.3389/fnsys.2023.1134594	Partial
6	van Praag H, Kempermann G, Gage FH. <i>Nat Rev Neurosci</i> 2000;1:191–198. doi:10.1038/35044558	Verified
7	Nithianantharajah J, Hannan AJ. <i>Nat Rev Neurosci</i> 2006;7:697–709. doi:10.1038/nrn1970	Verified
8	Galsworthy MJ, et al. <i>Behav Genet</i> 2005;35:675–692. doi:10.1007/s10519-005-3423-9	Verified
9	Matzel LD, Han YR, Grossman H, et al. <i>J Neurosci</i> 2003 (general learning ability in mice)	Partial
10	Gibson EM, Purger D, Mount CW, et al. <i>Science</i> 2014;344:1252304 (mouse optogenetic premotor-cortex stimulation promotes oligodendrogenesis and myelination)	Verified
11	Birrell JM, Brown VJ. <i>J Neurosci</i> 2000;20(11):4320–4324; Rabinovitch MS, Rosvold HE. <i>Can J Psychol</i> 1951;5(3):122–128	Birrell verified; Rabinovitch partial
12	Kraemer HC, et al. <i>Arch Gen Psychiatry</i> 2006;63:484–489. doi:10.1001/archpsyc.63.5.484	Verified
13	Leon AC, Davis LL, Kraemer HC. <i>J Psychiatr Res</i> 2011;45:626–629. doi:10.1016/j.jpsychires.2010.10.008	Verified
14	Percie du Sert N, et al. <i>PLoS Biol</i> 2020;18:e3000410. doi:10.1371/journal.pbio.3000410	Verified
15	Enrichment meta-analysis (rodent cognition $g \approx 0.44$, 95% CI 0.25–0.62): Univ. at Albany dissertation (scholarsarchive.library.albany.edu/etd/371); Léger M, et al. 2014 (duration of enrichment effects in mice)	Partial; not peer-reviewed
16	Fortin D, Rom E, Sun H, Yayon A, Bansal R. Distinct FGF/FGF receptor signaling pairs initiate diverse cellular responses in the oligodendrocyte lineage. <i>J Neurosci</i> 2005;25:7470–7479. doi:10.1523/JNEUROSCI.2120-05.2005	Verified
17	Tait DS, Bowman EM, Neuwirth LS, Brown VJ. Assessment of intradimensional/extradimensional attentional set-shifting in rats. <i>Neurosci Biobehav Rev</i> 2018;89:72–84	Verified
18	Browne RH. On the use of a pilot sample for sample size determination. <i>Stat Med</i> 1995;14:1933–1940	Verified (title to check)
19	Gallagher M, Burwell R, Burchinal M. Severity of spatial learning impairment in aging: development of a learning index for performance in the Morris water maze. <i>Behav Neurosci</i> 1993;107:618–626. doi:10.1037/0735-7044.107.4.618	Verified

Not found: any study of FGF17/FGF-family treatment improving learning in healthy young rodents; any study combining 7,8-DHF with enrichment; rat ICV FGF17 doses; Iram 2022 delivered dose; Indian prices for Alzet pumps and recombinant FGF17; a peer-reviewed validation of a Hebb-Williams subset; per-diem housing cost.